Chapter 1: How the Human Mind Works

Picture this. There is a device that could record every memory, idea, and dream in your brain and then send the entire contents to another person. This sounds game-changing, right? The truth is, we already have this type of technology; it is known as effective storytelling. The life of the human revolves around the ability to share experiences and information.

Being able to communicate well is all a matter of getting “in sync” with other people. You may have even noticed this. Maybe you have seen where people start matching steps as they walk, or how they imitate the gestures of others as they talk, or use grammar or phrases that the other person uses. But this syncing up doesn’t just happen in people’s mannerism or speech. It also happens in parts of the brain. This is what is known as “neural coupling,” and scientists believe it is an extremely important part of communication.

In one study on the functions of the brain, while communicating, scientists had a person tell a 15-minute story while also having the brain scanned with an MRI to record all of the activity in the different regions. They took this recorded and played it for 11 volunteers while they were having an MRI. As they listened to the recorded, their brain activity matched up with the storytellers.

This is a great way of looking at the brain during communication, especially considering there are quite a few studies that look at “the brain.” This doesn’t give us a good idea of what the brain does during communication because it takes more than one person to have a conversation. It also does a good job of showing that listening and talking aren’t two separate activities. The “neural coupling” takes place in the “production” and “comprehension” parts of the listener’s brain.

What’s even more interesting is that the firing in the brain regions happened first in the listener. That means the listener’s brain is acting in anticipation of what the speaker is going to say. They were priming their self, so to speak, as what they were expecting to hear. The more accurate these “predictive anticipatory responses” were in the listener, the better they were able to understand the story. This means that if what the speaker says goes

completely against what the person was expecting, the listener is going to be less likely to understand what is happening or they could just stop listening altogether.

In the 1930s, Sir Frederic Charles Bartlett performed some interesting experiments. Through these experiments, we have learned that listeners will "fill in" details about the things they are hearing. We also know that a person's memory of the speaker doesn't distinguish between the things they heard and what their brain told them.

In Bartlett's studies, he had students read a folktale and then asked them to retell the story. They would end up adding some details. For example, one part of the story said, "That Indian has been hit," and some of the students would say that an Indian had been hit by an arrow, or that the Indian had been killed. They would also change some of the other unfamiliar facts. You can even test this on yourself. Pick a short story that you don't know and read it. Then, in a couple of days, record yourself retelling the story out loud. See how the two compare.

The people involved in the story felt certain that their memories of the story were correct. They weren't able to tell the difference in their retelling. Why? Bartlett came to the conclusion that the mind understands things through "schemas." These are mental maps that relate objects and actions to one another. Once they are learned, the scheme works kind of like a mad lib book or a fill in the blank test. Once a person knows that the story is about Indians and canoes, their mind is going to "fill in" moccasins and arrows even if they weren't an original part of the story. The same can be true if you were to throw in something about samurai swords; they are going to leave out that part because it doesn't belong.

In the MRI study, performed at Princeton University, they discovered two neural mechanisms. The first is during communication, sound waves coming from the speaker, couples the listener's brain response with the brain response of the speaker. Second, the brain has created a common neural protocol that gives us the ability to use brain coupling to share information with others.

They did a second study where they took people into an fMRI scanner and scanned their brains as they were listening to or telling a real story. They looked at the similarity of the neural responses for the listeners in their

auditory cortices, which is the area of the brain that process sounds. They looked at the brain scans before the stories started to get a baseline reading. At that point, the brains were all very different and not synced up.

However, once the stories began, they noticed something amazing. All of the listeners' neural responses started to match up. They started to move in the same frequency to each other.

This is what scientists refer to as "neural entrainment," which refers to the process of the brain response to lock and align with the sounds of speech. But what drives this brain activity; the ideas that the speaker is conveying, or the sounds they are producing?

They continued on with their experiment. They took recorded stories and played them backward for the listeners. This kept much of the same auditory sounds, but it got rid of the meaning. They found that this created entrainment or the neural responses of every listener but didn't go any further into the other areas of the brain. Based on this, they figured that the auditory cortex would be entrained by sounds, regardless if there was an intelligible meaning or not.

Then they tried scrambling the words. This made the words comprehensible, but it sounded like a bunch of unconnected words. These words created alignment in the early language parts of the brains, but nowhere else.

They then formed the words back into sentences. While each sentence made individual sense, they don't go together to create a story. When this version was played for the listener, they start the entrainment move to every language area that processes grammatically coherent sentences. But once they finally played the full engaging story for them, they entrainment spread through more of the brain and created aligned responses between all of the listeners, which included the parietal and frontal cortices.

This led them to believe that the high-order cortical areas would become entrained to the ideas that a person shared as they placed sentences into an understandable narrative. If this conclusion should prove to be true, then if they shared a story to two listeners that were the same but used different sets of words, their brain's response would remain similar. To test this, they took the story they had been using and translated it into Russian.

They played English version for their English listeners, and then played the Russian version to Russian listeners and compared the neural responses. They didn't get similar responses, which they expected because the different languages sounded different. However, they did find that the responses within the higher-order areas were similar among all of them. This was because they all had the same understanding of the story.

Now that they had this information about what happens in a listener's brain, they wanted to look at what happened in the speaker's brain. The storyteller then underwent an MRI scan and they compared what happened within his brain to the listener's brain. Producing and comprehending speech are two different things. But amazingly, as mentioned earlier, the brain responses were very similar in the two groups.

The more understood the storyteller was, the more connected their brain waves were. This moved them onto another question. How can this coupling help us transmit a memory from one brain to another? They tested this by having people watch the pilot episode of *Sherlock* as they scanned their brains. Then, while still being scanned, they told the episode in the form of a story to another person.

In this episode, there is a scene where Sherlock takes a cab that is driven by the murderer. The subjects who were watching the scene showed specific neural patterns in the high-order brain areas. What was interesting is when they recalled this scene and shared it in story form to a person; the same patterns were seen in the listener's brain. This means that these brain patterns happen even when a person shares just a memory, not a real experience. This shows the important role language plays in sharing memories to other people.

The act of communication, though, is very far from perfect, and in many cases, we tend to fail to communicate in an effective manner, or are simply misunderstood. People can interpret the same stories in different ways. They decided to study this problem as well. They used the story "Pretty Mouth and Green My Eyes," by J.D. Salinger. In the story, the husband loses track of his wife during a party. He has to call a friend and asks, "Did you see my wife?" They took half of their listeners and told them that the wife was having an affair and told the other half that she was loyal.

What was interesting was that this simple sentence that they told the listeners beforehand was enough to cause their brain responses to be different. Both groups had similar brain responses to everybody in the group, but were different from the other group.

This shows that there is a lot of implications in communication, considering a single sentence was enough to affect brain responses. Good communication greatly depends on the listener and speaker having common ground. The problem with communication today is that most of us live a life where we are exposed to the same perspective each day. This doesn't give us much room for common ground amongst people who are different. If we lose our common ground completely, what would be able to do to ever communicate effectively?

One way to do this is to start having real conversations with each other were actually listen to each other's viewpoints. This gives us a chance to figure out a mutual understanding.

Mental Health and Communication

We are all humans, and as such, we all have our own problems. As much as we hate to think about it, our mental health plays into how we communicate. Stress levels, especially within families and job relationships, can create communication problems. This, and other mental health disorders can cause a person to misread or withdraw in social situations.

Whether you are the one struggling or a person you are talking to is, the most common signs are:

- Misinterpreting or missing simply social cues, like facial expressions, which causes misunderstandings.
- Make unreasonable demands, or they show a lack of concern because they are preoccupied with their own anxieties or fears.
- Perceive people inaccurately, which can cause paranoia or anxiety.
- Behave unpredictably, feel irritable, or have angry outbursts because of instability.
- Withdraw or remain silent when feeling depressed.

The only way around these problems, if you are the one suffering from them, is to seek help. Now, sometimes a little time to calm down or relax can help, especially if it has been a long and stressful day. But if you feel this way constantly, then seeking assistance from a professional is the best way to proceed.

Now, if you have to converse with somebody who you know is struggling with mental health issues, or is clearly stressed, there are a few things you can do. First, be brief and get to the point. The last thing a stressed person wants to hear is a longwinded story. If you are sharing personal feelings about something, use “I” statements so that you don’t come off as confrontational.

Most importantly, you can only speak for yourself. You don’t know how a person truly feels based solely on how they act. It is likely only going to make them more upset or stressed if they feel like you are speaking for them.

Alright, with all of this in mind, let’s move into the communication levels.